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Unsupervised density-based method for analyzing ion mobility in crystalline solid-state electrolytes

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Understanding the connection between chemistry, structure, and ion migration in solid-state electrolytes (SSEs) is vital to enable safer, more efficient all-solid-state batteries and other advanced electrochemical devices. Atomistic simulations offer invaluable access to the intricacies of this connection. However, extracting meaningful insights often requires detailed crystallographic knowledge and painstaking examination of simulation trajectories. This work introduces a density-based unsupervised method that accurately and efficiently identifies and categorizes crystallographic sites while providing a robust framework for analyzing ionic transport from *ab initio* and classic molecular dynamics simulations. Unlike previous schemes, it needs no prior structural knowledge and is adaptable to various material systems. Our approach, implemented in the open-source CrySF package and validated on representative SSEs such as $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$ (a garnet), $\text{Li}_{10}\text{GeP}_2\text{S}_{12}$ (a sulfide), and $\text{Li}_6\text{PS}_5\text{Br}$ (an argyrodite), effectively analyzes the interplay between structure, ionic mobility and collective migration phenomena, offering a powerful tool to accelerate the development of high-performance SSEs.

Solid-state electrolytes (SSEs) present a compelling alternative to traditional organic electrolytes in lithium-ion batteries. Non-toxic and non-flammable, SSEs not only enhance safety but also potentially increase energy density by enabling the use of pure lithium anodes^{1,2}. Intensive research is underway on a variety of structural families, including perovskites, sulfides, LISICON-like materials, argyrodites, and garnets, to find the most suitable material for battery integration^{3,4}. Several potential candidates demonstrating fast ionic conductivity (IC) ($\sim 10^{-3} \text{ S.cm}^{-1}$) have been identified to date, and progress in rationally optimizing their performance without compromising their chemical or mechanical stability requires a fundamental understanding of the mechanisms underlying their high IC⁵.

In inorganic SSEs, diffusion is often best described as a series of consecutive jumps between well-defined crystallographic sites, where mobile ions reside until they overcome the energy barrier to traverse the interstitial space to an adjacent site. Thus, accurately describing the site network geometry and connectivity is an essential step for characterizing and formulating strategies to enhance ionic mobility in SSE materials within specific engineering constraints. Atomistic simulation techniques, such as *ab initio* molecular dynamics (AIMD) and classical molecular dynamics (CMD), have significantly contributed to this task⁶. In combination with structural and electrochemical spectroscopy methods, these simulation schemes have enabled the detailed description of ionic diffusion pathways

along specific sublattices within the host material^{7,8}. However, external factors such as temperature, pressure, and aliovalent doping can significantly perturb the location, geometry, and connectivity of these sites. Moreover, exceedingly fast ionic transitions may render the precise identification and classification of crystallographic sites a challenging task from both an experimental and computational standpoint.

A closely related problem, highly dependent on the correct identification of diffusion pathways, is that of unraveling the impact of collective motion on the IC. Traditionally, the collective motion of ions has been described through van Hove correlation functions^{9,10}, which are limited to pair correlations. This limited perspective fails to capture the concentration effects and multi-ion interactions, which have been theoretically recognized as key elements in systems with high concentrations of mobile ions, such as SSE materials^{7,11}. While the cooperative diffusion of three or more atoms has been shown to enhance the IC by lowering the activation energy⁷, identifying and classifying these diffusion “strings” from long simulation trajectories efficiently and accurately remains a difficult task.

In light of these challenges, the development of advanced computational tools capable of analyzing atomistic trajectories with minimal supervision has emerged as a critical area of focus in materials science^{12,13}. This necessity has become more acute, considering that ever-increasing computer power and GPU-acceleration allow for the simultaneous

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execution of large simulation runs. Among the critical tasks that such tools should perform when considering SSEs are (i) the identification and characterization of the crystallographic network that mobile ions occupy, and (ii) the characterization of individual and collective ionic transition events between sites. Efficiently performing such analysis within an implementation that can be readily incorporated into a larger computational pipeline could significantly contribute to accelerating the rational development of high-performance SSE materials.

To address the complexity of analyzing ionic transport and its relation to material structure from CMD and AIMD trajectories, several schemes that require a priori knowledge of crystallographic sites location, geometry, and associated cutoff values have been reported^{14–17}. For instance, distance-based methods rely on predefined equilibrium positions of mobile ions and cutoff distances to determine site occupancy. However, these methods are sensitive to structural details and thermal noise, impacting their precision^{14,15,18}. Topology-based methods, such as convex-hull analysis¹⁶, are suitable for dealing with irregular site geometries and reducing the influence of thermal noise, although they still rely on prior knowledge of site polyhedra. Similarly, k-means-based methods use predefined equilibrium positions and k-means clustering to assign mobile ions to specific sites at each time step¹⁷.

Recently, unsupervised methods requiring fewer parameters have been developed^{11,19}. The “landmark method” is one such approach, in which the positions of mobile ions are transformed into a simplified “landmark space” based on local geometric features, followed by clustering to identify discrete ion sites. While this method is highly effective for site identification, it requires high-frequency sampling to achieve accurate results, which can be computationally expensive. Besides, it does not address the characterization of collective diffusion processes¹⁹. In contrast, a recent unsupervised method exploits the k-means algorithm to identify and categorize ionic jumps and diffusion paths¹¹. This technique bypasses the identification of crystallographic sites and operates with equilibrium and metastable positions, hindering the precision of ionic jump identification between adjacent sites.

Despite their contributions, each of the methods above presents trade-offs in terms of precision, flexibility, and applicability across different material systems. Density-based methods offer an alternative by defining sites through regions of high ion density separated by low-density areas, using specific thresholds. While this approach eliminates the need for material-specific information, it faces the challenges of determining a universal density threshold that reliably distinguishes closely spaced or rapidly ion exchanging sites^{20–22}.

In this work, we introduce a novel density-based, unsupervised method for identifying and categorizing crystallographic sites and analyzing atomistic trajectories. This approach addresses key limitations of previous methods, providing a comprehensive tool for studying ionic diffusion in SSEs. A detailed analysis was performed on three types of systems: tetragonal and cubic $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$, $\text{Li}_{10}\text{GeP}_2\text{S}_{12}$, and ordered/disordered $\text{Li}_6\text{PS}_3\text{Br}$ and Li_3PS_4 . In addition, only the site analysis was conducted for several relevant SSE materials: Li_3PS_4 , Li_3OCl , LiGaO_2 , $\text{LiMn}(\text{HCO}_2)_3$, and $\text{Li}_2\text{B}_{12}\text{H}_{12}$. Results align well with experimental and computational data, demonstrating the method’s accuracy and adaptability.

Results

Identification and classification of crystallographic sites

The scheme in Fig. 1 outlines the proposed workflow for identifying crystallographic sites from an atomistic trajectory. The process starts by converting atomic positions in a simulation trajectory into a voxel-based density map (details on voxel size selection are provided in Supporting Information section S1.1). Initially, the map includes all points for which the density, ρ , is greater than zero (Fig. 1a). Thus, it displays continually interconnected regions due to the high mobility of diffusive atoms in high-IC materials, making accurate identification of crystallographic sites impractical. To circumvent this, the map undergoes a filtering process in which only regions with a density above a specified threshold, ρ_{cut} , are retained. This procedure

seeks to eliminate the interstitial regions between crystallographic sites, which are less visited (i.e., have low values of ρ) because hopping events between adjacent sites are significantly less frequent than atomic vibrations within a single site. It is important to note that the distinction between a site and a transition region depends on density contrast. Sites that are weakly sampled, with densities similar to surrounding transitions, may not meet the criteria for identification and could remain undetected. Detection and occupancy determination of rarely visited sites may be improved by enhancing sampling in the simulation stage (e.g., by employing hybrid Monte Carlo - MD methods^{23–25}).

After applying the density threshold ρ_{cut} , the density map is spatially clustered using pDBSCAN, a modified version of the DBSCAN algorithm²⁶ that incorporates periodic boundary conditions²² (Fig. 1b). pDBSCAN produces m spacial clusters, C_i^s ($i = 0, \dots, m - 1$), each of which represents a distinct crystallographic site. To classify the generated type of sites (e.g., to distinguish between octahedrally and tetrahedrally coordinated sites), Principal Component Analysis (PCA) is applied to each C_i^s and the variances (λ_1, λ_2 , and λ_3) associated with the three principal components (PC_1 , PC_2 , and PC_3) are extracted. Through this procedure, we map each C_i^s to a single point in three-dimensional λ -space (Fig. 1c). These points are then clustered in λ -space using the k-means algorithm as implemented in scikit-learn²⁷. The optimal number of clusters, k , is determined by maximizing the silhouette coefficient s_k for each value of k between 2 and 10, as s_k is defined only for $k \geq 2$ and no reported crystalline SSE is known to contain more than 10 distinct types of crystallographic sites. If the maximum s_k is below a reasonable threshold value (defined here as 0.4), k is set to 1 (Fig. 1d). The clusters obtained through this process, $\{C_j^s\}_{j=0, \dots, k-1}$, define groups of crystallographic sites sharing similar volume and shape characteristics. These groups are termed “site types” and are labeled sequentially as type 0, type 1, ..., type $\leq Z$ ($Z \leq 9$). It is important to note that no crystallographic labels (e.g., Wyckoff positions) are assigned at this stage. Instead, site types are defined purely based on geometric similarity. When experimental structure refinements are available, the resulting classification can be directly compared to known crystallographic polyhedra.

Finally, each C_i^s must meet the criterion that its volume, $V(C_i^s)$ (i.e., the number of elements it contains times V_{voxel} where $V_{\text{voxel}} = (v_{\text{size}})^3$ and v_{size} is the voxel edge length), falls within a specified range: greater than V_{min} and less than V_{max} (Fig. 1e). We have selected $V_{\text{min}} = 0.28$ and $V_{\text{max}} = 3.0$, corresponding to the volumes of spheres with radii of 0.4 and 0.9 Å, respectively, for their effectiveness across various materials tested with this methodology. If $V(C_i^s)$ is outside this range, the density threshold ρ_{cut} is incrementally adjusted by an amount ρ_{step} ($\rho_{\text{step}} = 1/V_{\text{voxel}}$) until the condition is met. This process eliminates overly large sites that are fully interconnected (Fig. 1b, black arrows) and avoids identifying excessively small sites linked to interstitial regions. Once the correct ρ_{cut} is found, the crystallographic sites and their types are identified (Fig. 1f).

Using this methodology, it is straightforward to extract site occupancies, amplitudes, and volumes, as well as to perform nearest neighbor analysis (see Methods section). It is important to stress that this method does not consider the entire volume of anion polyhedra as a crystallographic site. Instead, a crystallographic site is defined by the volume that the center of mass of the mobile ions can actually visit. In that regard, the interstitial regions in this work do not correspond to the vertices or edges of intersecting polyhedra, but to volumetric regions in 3D space that may encompass small regions within these polyhedra.

Diffusion analysis

In addition to characterizing the crystal lattice, we introduce a methodology to efficiently and precisely track inter-site jumps of diffusing atoms. To this end, we store the trajectories of diffusing atoms in the $n \times p$ matrix Ω , where n is the number of atoms and p the number of output steps in the trajectory (Fig. 2). Thus, an entry $\Omega_{k,l}$ of Ω corresponds to the label of the site visited by atom k during step l rather than its 3D coordinates. With this structure, it is

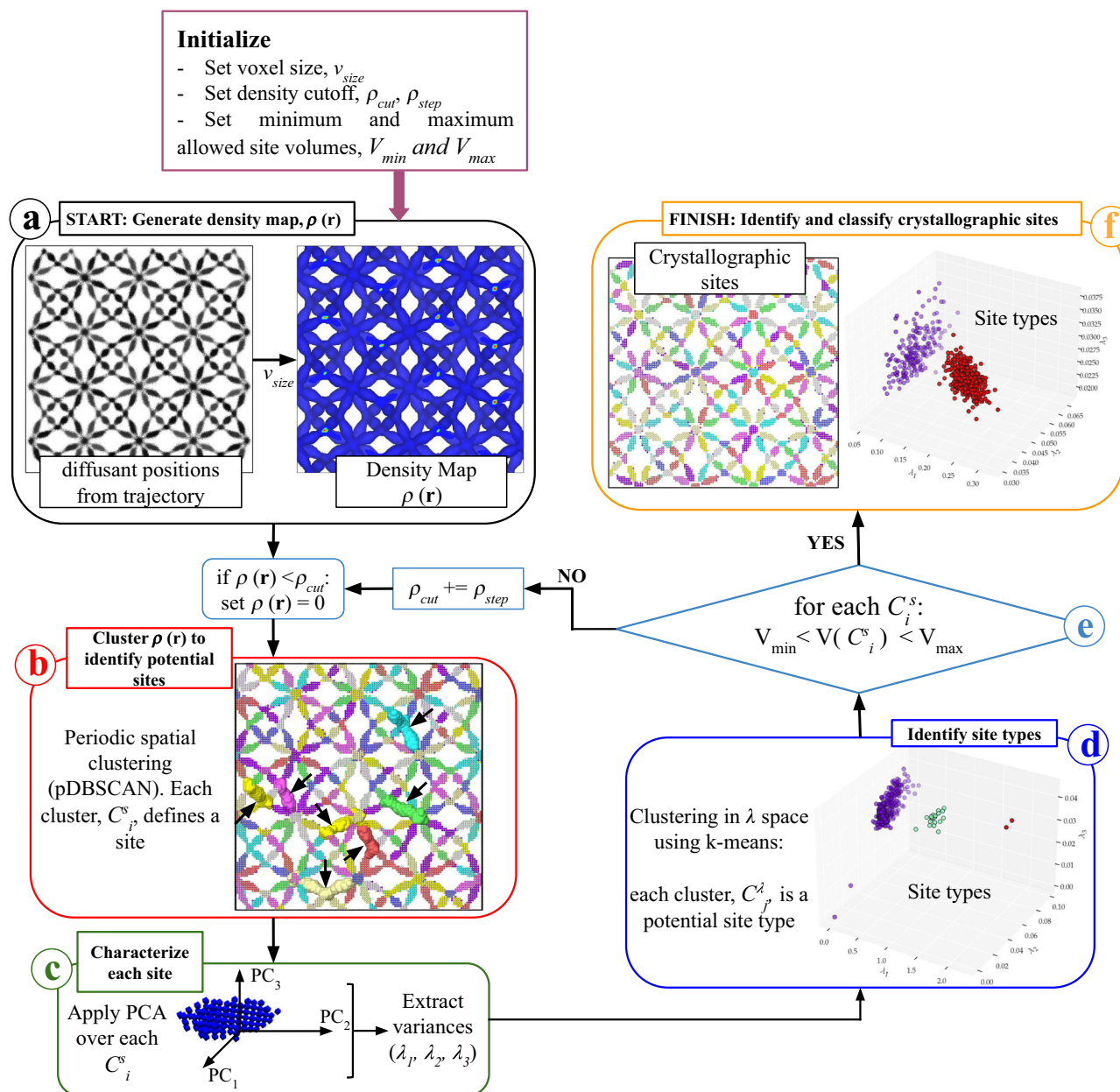


Fig. 1 | Workflow for crystallographic site identification in CrySF. Schematic workflow for identifying crystallographic sites from atomistic trajectories. **a** Generate a voxel-based density map from atomic positions in a simulation trajectory. Only regions with a density above a specified threshold, ρ_{cut} , are retained. **b** The map is spatially clustered using periodic DBSCAN (pDBSCAN) to identify potential crystallographic sites, C_i^s . **c** To characterize the potential sites, Principal Component Analysis (PCA) is applied to each C_i^s , and the variances ($\lambda_1, \lambda_2, \lambda_3$) from

the three principal components (PC_1, PC_2, PC_3) are extracted. **d** Variances are clustered in λ -space using k-means to identify site types. **e** Each C_i^s must meet the criterion that its volume, $V(C_i^s)$, falls between V_{min} and V_{max} . If $V(C_i^s)$ is outside this range, ρ_{cut} is incremented by ρ_{step} , and steps (**b–d**) are repeated until this condition is met. **f** Once the correct ρ_{cut} is found, the crystallographic sites and their types are identified.

straightforward to identify atomic jumps in Ω as changes in the site labels across consecutive steps. Specifically, a jump occurs whenever $\Omega_{k,l+1} - \Omega_{k,l} \neq 0$ (see Fig. 2, green arrows).

Consecutive reverse jumps, in which a diffusing atom reverses its path, can be readily identified in Ω (see Fig. 2, orange arrows). With this information, the jump frequency, ν_k , as well as the jump coefficient, α_k (a measure of network tortuosity), for atom k can be quantified as

$$\nu_k = J_k^T / t_T, \quad (1)$$

$$\alpha_k = J_k^R / J_k^T, \quad (2)$$

where J_k^T and J_k^R are the total number of jumps and reverse jumps for atom k , respectively, and t_T is the trajectory length.

Insightful descriptors of atomic collective motion can also be extracted from analyzing matrix Ω , such as the number of atoms jumping within a given time interval, Δt_j , where Δt_j represents an integer multiple of the time between output steps in the trajectory, Δt . For instance, Fig. 2 shows that, in each of two consecutive intervals of length $\Delta t_j = 3\Delta t$, several atoms (four in the first interval and three in the second one) jump simultaneously (including two reverse jumps in the second one). Clearly, an exceedingly short Δt_j may lead to no jump simultaneity, while a long one may show a large number of simultaneous jumps that bear little correlation. To balance resolution and physical relevance, we define Δt_j as a small multiple of the

average jump period, based on a heuristic criterion developed through empirical testing across different SSEs. This approach effectively captures short-timescale cooperative events while minimizing the risk of merging unrelated transitions (see section S1.2 of the Supporting Information for further discussion).

Likewise, we can identify strings of atoms involved in concerted migration processes²⁸. More precisely, atoms $a_1, a_2, \dots, a_n$ are defined to consecutively diffuse in a cooperative string between steps j and $j + 1$ if

$$\Omega_{a_i,j} = \Omega_{a_{i'},j+1}, \text{ for } i \neq i'. \quad (3)$$

A graphical example of how this condition is readily identified in matrix Ω is shown in Fig. 2 (right panel). In simple terms, this condition means that each time an atom moves to a new site, another atom occupies the vacated site in the immediately subsequent step. It is worth noting that these strings are identified independently for each pair of consecutive output frames and may not persist over time. Our method does not track the continuity of strings across multiple frames but instead captures localized, frame-to-frame cooperative events. Additionally, while the method explicitly identifies cooperative jumps, some strings may arise by chance in fast or densely populated systems. Future extensions could incorporate statistical baselines, such as randomized trajectory comparisons, to estimate the background frequency of such coincidental alignments, as suggested in prior work using Poisson statistics to distinguish correlated from uncorrelated motion²⁹.

To quantify these strings, we represent their occurrence in terms of frequency:

$$f_n = s_n/t_T, \quad (4)$$

where s_n is the total number of strings of length n . This representation accounts for variations in trajectory lengths, providing a consistent basis for comparing string frequencies across different simulations and materials.

The identification and characterization of crystallographic sites, as well as the diffusion analysis scheme described above, have been implemented in the Crystallography Sites Finder (CrySF) code (see Code Availability section). All the results below are automatically extracted from the simulation trajectories by our package. While CrySF supports both classical and AIMD simulations, the reliability of the analysis depends on standard trajectory quality criteria. Sufficient equilibration and trajectory length are necessary to ensure that all relevant sites are sampled, particularly in systems where the number of available sites exceeds the number of diffusing ions. Additionally, the frequency at which trajectory frames are saved plays a critical role. If the interval between output frames is too long, atomic jumps may go undetected or appear as simultaneous events. CrySF provides user-controlled parameters to adjust the time resolution used during post-processing.

In the following, we apply the proposed methodology to analyze the dynamics of diffusive atoms in three materials: tetragonal and cubic garnet $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$, sulfide $\text{Li}_{10}\text{GeP}_2\text{S}_{12}$, and argyrodite $\text{Li}_6\text{PS}_5\text{Br}$. The objective is to characterize the atomic diffusion network and gain insight into the collective behavior of diffusive atoms by calculating the relevant properties described above.

System 1: tetragonal and cubic garnet $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$ (LLZO)

The garnet LLZO is a solid electrolyte material, known for its high IC and stability against lithium metal⁴. LLZO has two polymorphs: a low-IC tetragonal phase ($I4_1/acd$, Space Group 142) and a high-IC cubic phase ($Ia\bar{3}d$, Space Group 230), stable above 400–625 K depending on impurity levels^{30,31}. The tetragonal phase is characterized by an ordered, fully occupied distribution of Li ions across three types of sites: tetrahedral (8a) and octahedral (16f, 32g) sites. In contrast, the cubic phase features a disordered, partially occupied distribution of Li ions across two types of sites: tetrahedral (24d) and octahedral (48g/96h)³².

We applied our methodology, as implemented in the CrySF code, to perform an unsupervised identification and characterization of Li sites in

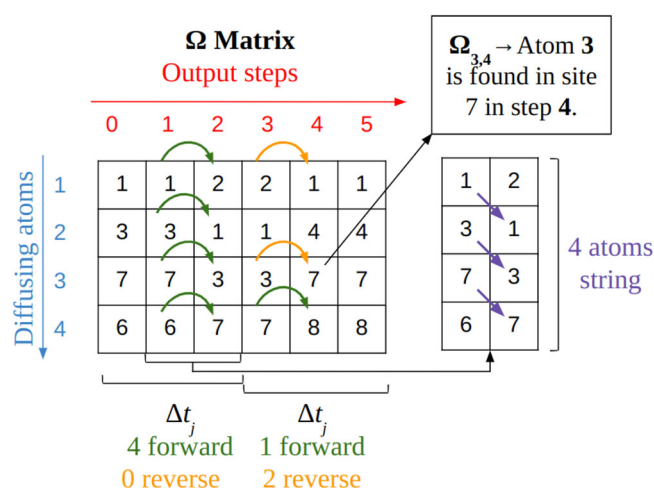


Fig. 2 | Matrix representation of atomic jumps and diffusion strings. Matrix Ω for a system of $n = 4$ diffusing atoms during $p = 6$ output steps. Green and orange arrows indicate forward and reverse atomic jumps between sites, respectively. The bottom panel details the number of atoms jumping (forward and reverse) in each $\Delta t_j = 3\Delta t$ time intervals. Purple arrows in the right panel highlight a group of $n = 4$ atoms forming a diffusion string, following the definition in eq. (3).

LLZO using CMD simulation data. Both the tetragonal and cubic phases of LLZO were analyzed (for simulation details see the Methods section). For the tetragonal phase, with a voxel size of 0.1 Å, our analysis successfully identified 64 type 0 sites, 256 type 1 sites, and 128 type 2 sites (Fig. 3a). These results align perfectly with the known Li positions: 8a (type 0), 32g (type 1), and 16f (type 2) within the 8-unit cells system. Each type displayed a consistent number of nearest neighbors (NN): 8a sites are surrounded by four 32g sites, 32g sites are connected to one 8a, one 32g, and two 16f sites, and 16f sites are linked to four 32g sites. The average occupancies (see Methods section, Eq. (7)) reached as high as 0.99, indicating a dense and regular arrangement of Li ions.

For the cubic phase and using a larger voxel size of 0.2 Å, our approach identified 192 type 0 sites and 384 type 1 sites, which correspond to the 24d and the combined 48g/96h Li sites within an 8-unit cells system (Fig. 3b). The nearest neighbor configurations in this phase are simpler than those in the tetragonal phase: 24d sites interact with four 48g/96h sites, and 48g/96h sites connect with two 24d sites. The estimated occupancies were 0.42 for 24d sites and 0.92 for 48g/96h sites.

In both phases, the total number of type sites (with 48g/96h treated as a single site)³³ as well as the number of nearest neighbors and their occupancies, are in good agreement with previous experimental and theoretical studies for both LLZO phases^{33–35}, validating our methodology. It is worth noting that, in the cubic phase, 48g/96h sites were merged (see inset in Fig. 3b). This is because both sites strongly overlap. Isolating 96h sites would require large ρ_{cut} values, leading to a loss of resolution in distinguishing interstitial spaces. The challenge of identifying 96h sites has been corroborated by previous studies employing density-based and landmark-basis methods, where evidence of 96h sites was absent²⁰, and the total number of identified sites was incomplete (83 out of 96)¹⁹.

The site amplitude, defined as the distance from the site centroid to the farthest voxel, is presented in Fig. 3c, d for the different types of Li sites in tetragonal and cubic LLZO, respectively. Notably, while several 32g (type 1) and 16f (type 2) sites in tetragonal LLZO display similar amplitudes (see Fig. 3c), our approach can effectively distinguish these two octahedral geometries. Such a distinction is crucial for understanding the pathways of Li diffusion in this material, highlighting our method's ability to resolve geometric nuances despite geometric similarity. Meanwhile, the wider site amplitudes of 24d (type 0) and 48g (type 1) sites in cubic LLZO (see Fig. 3d) compared to their tetrahedral and octahedral counterparts in tetragonal LLZO contribute to reduced interstitial regions between sites, resulting in a

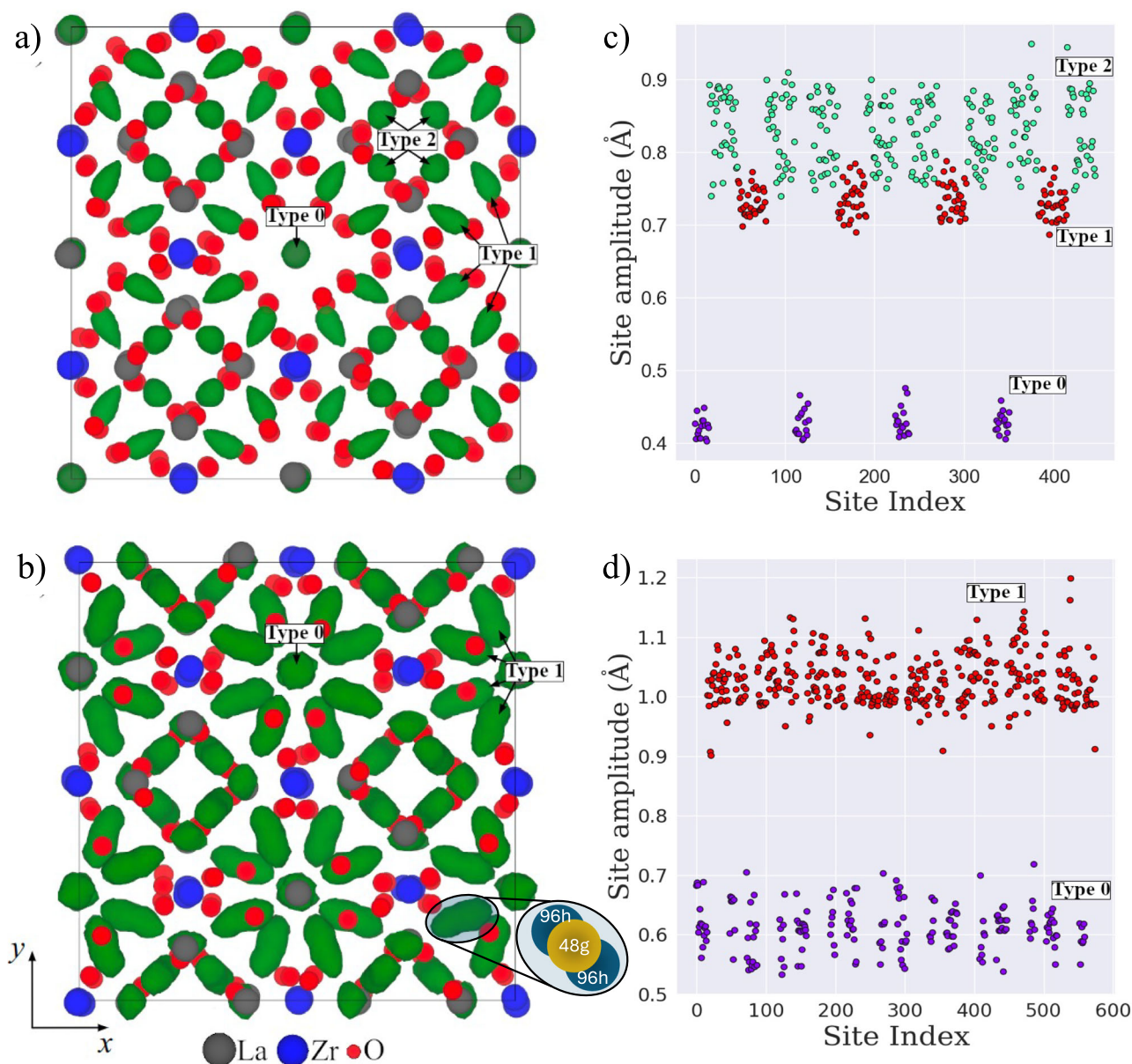


Fig. 3 | Crystal structures and Li site distributions in LLZO. Crystal structures of the unit cell for LLZO in **a** the tetragonal phase and **b** the cubic phase, illustrating the distinct lithium (Li) sites predicted by our methodology, represented as green

more interconnected Li diffusion network. This enhanced connectivity aligns with the higher Li-ion mobility observed in the cubic system, as previously suggested by Buschmann et al.³⁶

Figure 4a, b display the distribution of the number of distinct sites visited by lithium atoms for the tetragonal and cubic phases of LLZO, respectively. In the tetragonal phase, approximately 74% of the 448 Li atoms visited only one site throughout the entire simulation, indicating limited mobility. Only 26% of the atoms visited between two and five sites. In contrast, the highly conductive cubic phase exhibits a broad distribution of visited sites, with about 18% of Li atoms visiting up to eight different sites, and a small fraction of them visiting up to 16 sites in only 5 ns.

Ions may often perform reverse jumps in fast ionic conductors. If these ions are energetically or kinetically trapped by, for instance, low accessibility to vacancies, an encircling arrangement of repulsive dopants, or structural defects, the fraction of reverse jumps over the total number of jumps, α_k (Eq. (2)), may be quite significant. Fig. 4c shows α_k for tetragonal and cubic LLZO (atoms that do not jump are excluded, hence the gaps along the x-axis

density volumes. The amplitude distributions of the Li sites are depicted in **c** for tetragonal LLZO (Type 0 (8a), Type 1 (32g), and Type 3 (16f)) and in **d** for cubic LLZO (Type 0 (24d) and Type 1 (42g/96h)).

for the tetragonal system). In poorly conducting tetragonal LLZO, where no vacancies exist, atoms largely jump from one site to another concertedly. In such cases, $\alpha_k = 0$, as repulsion from an incoming atom tends to propel the atom occupying a given site forward. However, the low frequency of these jumps maintains the IC low. Conversely, the value of α_k for the cubic phase is on average 0.9, with the lowest and highest α_k values at 0.64 and 0.98, respectively. This surprising result indicates that diffusing atoms experience, on average, nine reverse jumps before moving forward. Those recording an α_k between 0.6 and 0.8 are exceptionally mobile (i.e., on the right tail of the distribution in Fig. 4b), while those with $\alpha_k \sim 0.98$ may have limited access to vacancies during the simulation run.

The frequent occurrence of reversible jumps in cubic LLZO is indicative of a lack of cooperative motion in atomic displacements. This is schematically illustrated in Fig. 4d: a Li atom in a tetrahedral site (type 0) will quickly jump to an adjacent vacant octahedral site (type 1). However, octahedral sites have a much higher occupancy (0.92) than tetrahedral sites (0.42), which means that, on average, slightly less than one vacant octahedral site surrounds each tetrahedral site. Thus, if the next tetrahedral site along

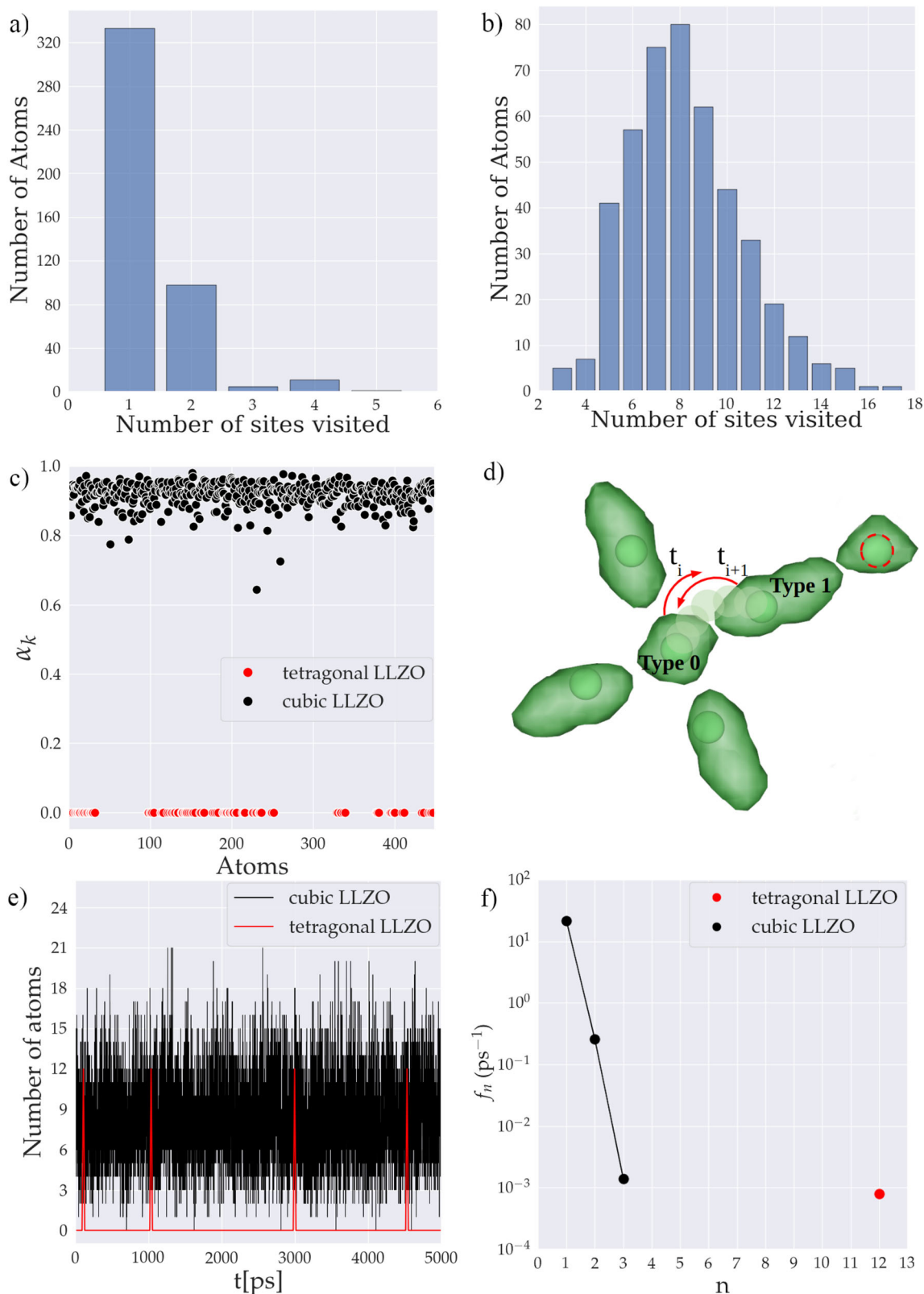
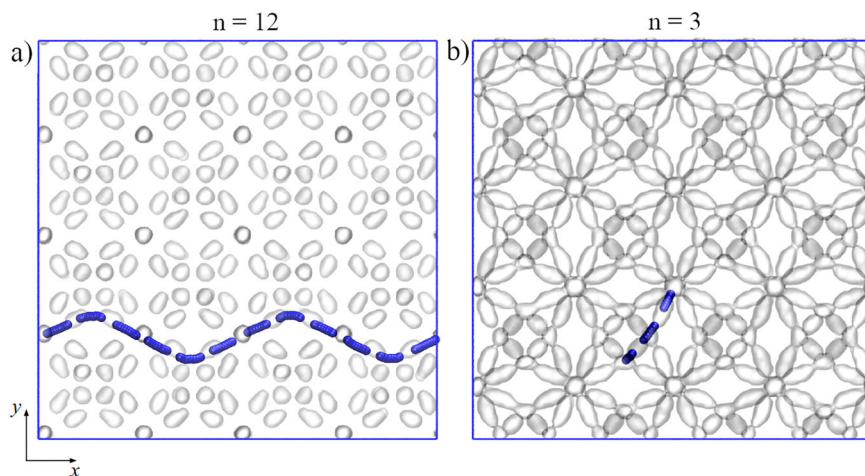


Fig. 4 | Li dynamics and diffusion strings in LLZO. Number of distinct sites visited in **a** the tetragonal and **b** cubic phases of LLZO. **c** Jump coefficient, α_k (Eq. (2)). **d** Schematic representation of reverse jumps in cubic LLZO: a Li atom moving from a type 0 to a type 1 site jumps back due to path blocking by a non-diffusive Li atom

(dashed red line). Consecutive jumps are denoted by t_i and t_{i+1} in **(d)**. **e** Number of atoms jumping simultaneously within time intervals $\Delta t_j = 31.5$ ps for tetragonal LLZO, and $\Delta t_j = 0.25$ ps for cubic LLZO. **f** String frequency, f_n , as a function of strings length n (Eq. (4)).

Fig. 5 | Longest diffusion strings in LLZO.

Representation of the longest atomic strings involved in the concerted diffusion process within **a** tetragonal ($n = 12$) and **b** cubic ($n = 3$) phases of LLZO.



the path is occupied by another Li (marked with the red dotted line), the moving Li atom will quickly reverse its trajectory. Considering that site occupancies are theoretically limited (0.88–1.0 for octahedral sites, and 0.33–0.55 for tetrahedral²²) it is probable that having the lowest possible octahedral occupancy is optimal for achieving the highest possible IC. This perspective aligns with our recent theoretical work²² and experimental findings by Adams et al.³⁷, challenging the notion that higher octahedral occupancies enhance IC^{38–40}.

Figure 4e illustrates the number of simultaneous jumps occurring within time spans of $\Delta t_j = 31.5$ and 0.25 ps for the tetragonal and cubic LLZO phases, respectively, throughout the simulation run. These time spans are determined based on the average jump period calculated by CrySF for these systems (see Supporting Information section S1.2 for the selection criteria). In the tetragonal phase, the large number of events involving exactly 12 Li atoms indicate the existence of closed diffusion loops comprising exactly 12 Li sites. In contrast, the cubic phase exhibits a wide distribution in the number of atoms participating in simultaneous jumps, with an average of approximately 9 atoms (2%) consistently engaging in this behavior. To unravel whether simultaneous jumps in cubic LLZO are correlated, we examine the length, n , and frequency, f_n (Eq. (4)), of concerted diffusion strings in Fig. 4f.

In the tetragonal phase, only strings of length $n = 12$ are present. Further analysis of the trajectory for the atoms involved in those strings reveals a diffusion pathway containing the sites 8a-32g-32g-8a-32g-32g per unit cell, which is in good agreement with recent theoretical studies³⁴ (Fig. 5a) and contradicting earlier CMD results⁴¹. This same diffusion pathway was observed by applying our methodology to AIMD simulations of tetragonal LLZO at 800 K (trajectory provided by Cazorla et al.⁴², see Supporting Information section S2.1). In contrast, the cubic phase exhibits string lengths ranging from $n = 1$ to 3 (a typical diffusion string is shown in Fig. 5b), with $n = 1$ having by far the largest frequency. This result suggests that most simultaneous jumps in the system are not spatially correlated (i.e., non-cooperative), which is likely associated with the abundance of vacant sites. From the values of α_k in Fig. 4c, we know that a significant fraction of these uncorrelated jumps correspond to atoms retracing their path (~90%). If reverse jumps are not considered, we can readily establish that at least ~19% of the jumps contributing to long-range diffusion must occur within strings of 2 or 3 atoms. This non-negligible fraction, supports the argument that concerted migration of ions is a crucial contributor to the high IC in cubic LLZO^{7,43}, as it tends to reduce the hopping barrier.

System 2: $\text{Li}_{10}\text{GeP}_2\text{S}_{12}$ (LGPS)

The sulfide LGPS is a superionic conductor that stands out due to its exceptional IC of up to $1.2 \times 10^{-2} \text{ Scm}^{-1}$ at room temperature⁴⁴, placing it at the forefront of solid electrolyte research. We analyze the crystallographic structure and diffusion pathways using the proposed scheme (Fig. 1), as

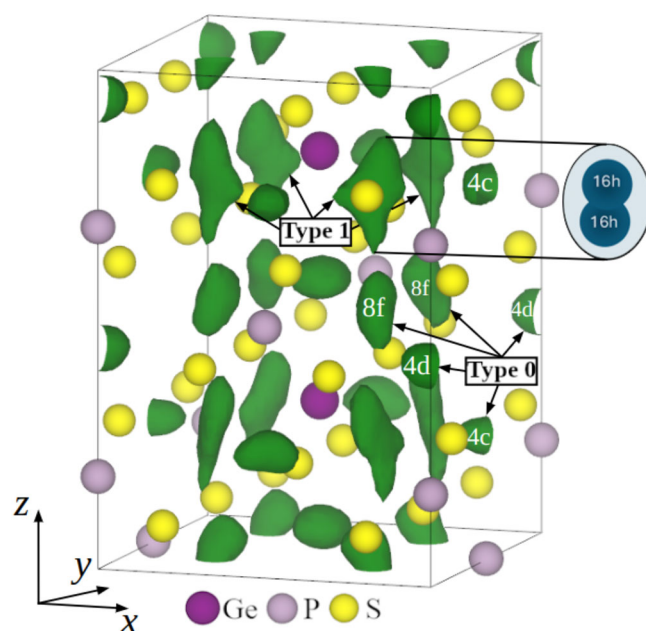


Fig. 6 | Crystal structure and Li sites in LGPS according to CrySF. Crystal structure of the unit cell of tetragonal LGPS, showing the distinct Li sites as green volumes.

implemented in the CrySF code, using an AIMD trajectory at 650 K (see Methods section) and with voxels size of 0.3 Å.

LGPS exhibits a tetragonal crystal structure within the $P4_2/nmc$ space group and includes four distinct lithium sites: 16h, 8f, 4d, and the more recently identified 4c⁴⁵. This structural arrangement enables unique ion transport dynamics, with the 16h and 8f sites facilitating one-dimensional (1D) pathways along the z -axis, and the 4d and 4c sites supporting two-dimensional (2D) mobility across the xy -plane⁴⁶. CrySF identified 24 sites, classified into 16 type 0 and 8 type 1 sites. Type 0 sites correspond to 4d, 4c, and 8f positions, while type 1 sites align with the 16h positions (Fig. 6). Notably, the rapid transition of Li atoms between pairs overlapping 16h sites causes these sites to be treated as a single site in our methodology, reducing the 16h sites to 8 type 1 sites (see inset Fig. 6), a phenomenon akin to the behavior observed with the 48g/96h sites in the cubic phase of LLZO.

Similar to the LLZO case, our methodology applied to LGPS demonstrates promise in the unsupervised identification of crystallographic sites occupied by diffusive atoms, successfully distinguishing between high-jumping-rate sites, such as 16h and 8f. However, LGPS presents unique challenges. Even the landmark-based method misidentifies 58 sites

compared to the 32 reported experimentally¹⁹, while our approach reduces these to 24 due to overlapping adjacent 16h sites being treated as a single site. Additionally, only two primary site types (type 0 and type 1) were identified instead of the expected four (4d, 4c, 8f, and 16h). This discrepancy in the number of types can be attributed to significant differences in the volume and shape between the merged 16h sites and the other sites (4d, 4c, and 8f), which affects the clustering results. Specifically, these structural variations led to a maximum s_k for $k = 2$ during the k-means clustering step in the algorithm scheme (see Fig. 1d).

Despite these challenges, our analysis consistently identifies four nearest-neighbor configurations across all sites (types 0 and 1), in agreement with experimental crystallographic data⁴⁵. Since CrySF distinguishes only between type 0 and type 1 sites, we assigned specific site types (4d, 4c, 8f, 16h) visually. Based on this classification, the average occupancies are 0.86 for 4d, 0.62 for 4c, 0.50 for 8f (i.e., 0.62 for type 0 sites), and 0.63 for 16h (i.e., 0.63 for type 1 sites). These results are consistent with experimental data at 100 K, which report occupancies of 0.89 for 4d, 0.77 for 4c, 0.72 for 8f, and 0.47 for 16h⁴⁵. However, notable discrepancies of approximately 19% for 4c and $\pm 30\%$ for 8f and 16h are observed. The reduced occupancy of 4c sites may arise from the distribution of Ge atoms known as “parallel to the z -axis”⁴⁷, which impacts 4c occupancy since Li diffusion predominantly occurs in the xy -plane via 4d sites⁴⁷. For 8f and 16h sites, the deviations can be attributed to spatial clustering (Fig. 1b), where proximity and high rate of jumps between these sites lead to an overestimation of the 16h volume and an underestimation of the 8f volume. Consequently, the calculated 16h occupancies appear higher, while 8f occupancies are lower than experimental values.

The diffusion analysis through CrySF (Fig. 7a) shows that approximately 40% of the Li atoms in the simulation cell visited between 12 and 14 distinct Li sites. Since the 1D channels along the z -axis comprise only four different Li sites (two 16h and two 8f), migration of Li atoms to adjacent 1D channels through interconnecting 4d and 4c sites must necessarily occur. Indeed, neutron scattering data by Weber et al.⁴⁶ indicated that Li atoms exhibit quasi-isotropic diffusion in LGPS from one-dimensional diffusion channel crossing (i.e., interchannel migration), a phenomenon also reported in phosphate cathode materials^{48,49}.

Figure 7b indicates that 60–80 % of all jumps were followed by a reverse jump. Since 1D diffusion is the most typical transport mechanism, we can explain this observation by considering that an atom must wait for the following site along its path to be vacated, leading to a high probability of path reversal (see inset A Fig. 7c). Therefore, any reduction of α_k by, for instance, enhancing cooperative diffusion or facilitating interchannel migration, could significantly boost the IC.

Simultaneous jumps occurring within intervals of 0.08 ps are presented in Fig. 7c. Over 90% of simultaneous jumps involve between one and three atoms, suggesting that they largely occur within the 1D channels formed by four Li sites. In this regard, interchannel hopping could be a crucial mechanism for Li diffusion in LGPS, as the vacancy generated by a Li atom crossing channels may trigger a collective diffusion response (see inset B Fig. 7c), as previously suggested in other materials^{48,49}. To verify whether simultaneous jumps are the result of cooperative behavior, Fig. 7d depicts CrySF estimation of the frequency of diffusion strings in LGPS. We observe that diffusion strings of $n = 2$ to 4 occur with relatively high frequency ($n = 1$ corresponds to the trivial string of one atom) and, if reverse jumps are not taken into account, we can readily conclude that at least 20 % of the jumps contributing to long-range diffusion should occur within a cooperative string (a snapshot of $n = 3$ and $n = 4$ strings are depicted in Fig. 7e). This result corroborates previous reports highlighting the contribution of concerted motion in 1D channels as a key factor in the superionic conductivity of LGPS⁷. As in the case of the diffusion strings in tetragonal LLZO (Fig. 5a), the limit $n = 4$ is given by the size of the simulation box. Therefore, diffusive strings of $n \geq 4$ atoms can occur, with a period of at least ~ 53 ps.

System 3: Li₆PS₅Br (LPSB)

Li-argyrodites Li₆PS₅X (X = Cl, Br, I) represent a relatively new and promising class of superionic conductors, characterized by their high IC at

room temperature (exceeding 10^{-3} S cm⁻¹)⁵⁰. These materials have a cubic crystal structure in the F43m space group, where Li atoms form pseudo-octahedral cages around S²⁻ sites, with four cages per unit cell. In this study, we focus on the case where X = Br, as a representative of the Li-argyrodites family and consider both ordered and disordered LPSB systems^{50,51}.

We first applied CrySF to 200 ps AIMD trajectories of ordered and disordered LPSB systems at 650 K (see Methods section), with a voxel size of 0.2 Å. In LPSB, Li atoms occupy the 48h/24g sites, commonly denoted as T5/T5a. Additionally, recent studies have uncovered the presence of a 48h site type, referred to as T2, located at the corners of the cage. Using the methodology outlined in Fig. 1, we identified 24 sites classified as type 0. These sites correspond to clusters of T5-T5a-T5 sites, forming four octahedral cages with six sites per cage (Fig. 8 left). As previously observed for cubic LLZO and LGPS, adjacent sites with high jumping rates merge into a single site. In ordered LPSB, the crystallographic T5-T5a-T5 sites coalesce into a larger site (type 0) due to the high frequency of the so-called doublet jumps (Fig. 8 right)^{52,53}. Each type 0 site has 4 nearest neighbors, and the computed occupancy reaches 0.91, close to the refined value of ~ 1.0 ⁵³. In general, minor inaccuracies in the evaluation of occupancy may result from transient site-to-site jumps, during which atoms briefly occupy low-density transition regions (see inset of Fig. 9c). Such errors can be leveled out by increasing the density tolerance during the site detection.

By lowering the V_{min} threshold (see Fig. 1), CrySF was able to identify a second type of site associated with smaller T2-type sites. However, due to the low occupancy of T2 sites, estimated as approximately 0.06 in disordered and non-stoichiometric Li-argyrodites⁵⁴, and not reported at all in some stoichiometric cases⁵³, only three T2 sites of the 48 that should be present in the simulation box were identified, with an average occupancy of 0.02 (see Supporting Information section S2.2). It may be possible that, in longer trajectories, the number and occupancy of T2 sites would increase. However, in the remainder of our analysis, we will focus exclusively on type 0 sites, as these are the most predominantly visited in this system.

Figure 9a shows that each Li atom visited six unique Li sites within the ordered LPSB structure. This finding suggests that, during the simulation time, Li atoms are primarily engaged in intracage jumps, specifically between T5 $\rightarrow$ T5 sites (Fig. 8 left). As a result, each Li atom was able to fully explore and occupy every site within the octahedral cage in which it was originally contained. This result is consistent with previous reports of low IC in ordered Li-argyrodites, where no intercage jumps were observed^{52,55,56}. Moreover, Fig. 9b shows an average α_k of 0.31, indicating a low rate of subsequent reverse jumps and suggesting that Li atoms move in a preferential direction within the cages.

According to CrySF, simultaneous jumps of 1 to 4 atoms are common (the selected Δt_j was 0.06 ps) (Fig. 9c). 3 and 4 atoms jumping simultaneously originate from pathways such as those marked with the yellow and red dashed lines along the cage edges in Fig. 8 left. The concerted jump of three Li atoms around one face of the octahedral cage has been previously reported and is known as trigonal-prismatic reorganization, while the concerted motion of four Li atoms along the center plane of the cage is defined as a rigid rotation⁵². However, simultaneous jumps of 1 and 2 atoms, as observed in our analysis, should not exist in a vacancy-free system. This result arises from atoms temporarily occupying interstitial regions between sites, allowing the initial jump to proceed while the displaced atom resides momentarily outside any defined site (see inset of Fig. 9c). Notably, these interstitial regions, though visited only during a transition attempt, would be part of T2 Li sites.

The strings of jumping atoms with lengths ranging from $n = 1$ to $n = 4$ identified by CrySF were consistent with the discussion above (Fig. 9d). For $n = 3$ strings, two types of concerted movements were observed. In the first type, a string of ions moving along the center-plane pathway of the cage containing 4 sites (red dash-line in Fig. 8 left) either starts or ends at an interstitial region (see snapshot in Fig. 9e left panel. Red circle shows interstitial site). The second, less frequent case, corresponds to trigonal-prismatic reorganization (Fig. 9e middle panel)⁵². Finally, strings of length $n = 4$ are the result of rigid rotations⁵² (Fig. 9e right panel).

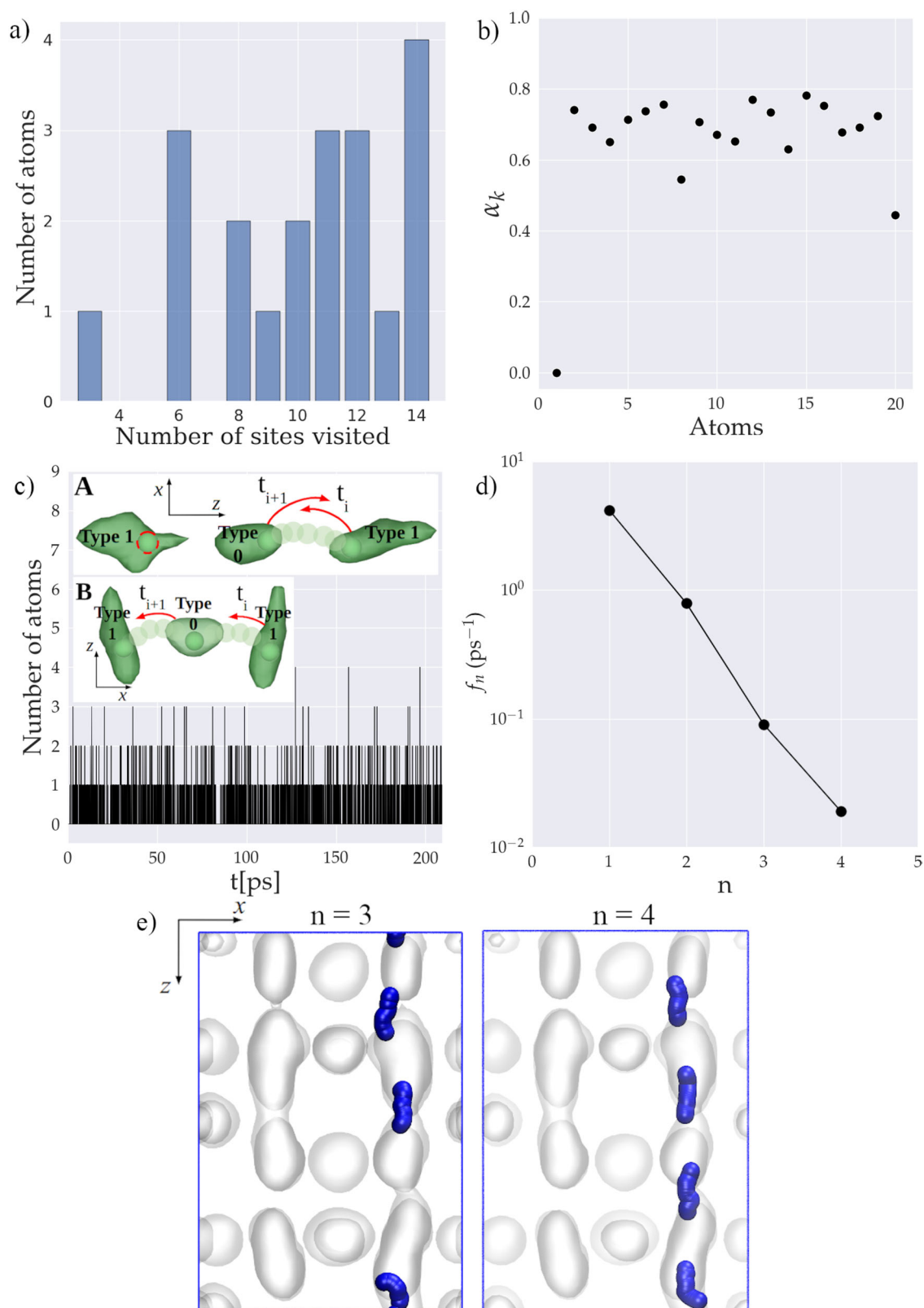
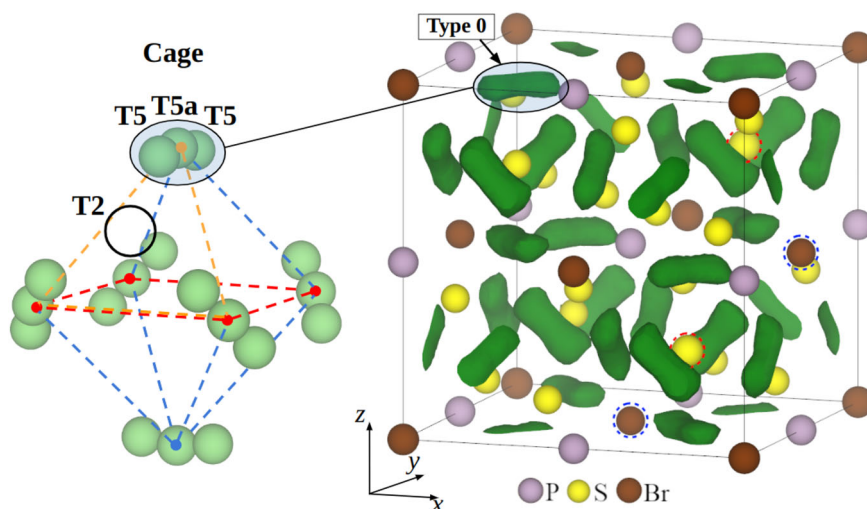


Fig. 7 | Li dynamics and diffusion strings in LGPS. **a** Number of distinct sites visited. **b** Jump coefficient, α_k (Eq. (2)). **c** Number of atoms jumping simultaneously within $\Delta t_j = 0.08$ ps. Inset A: Illustration of reverse jumps in 1D channels (oriented along the z axis) caused by path blocking from a non-diffusive Li atom (dashed red

line). Inset B: Representation of interchannel jumping between adjacent channels. Consecutive jumps are denoted by t_i and t_{i+1} in insets A and B. **d** String frequency, f_n , as a function of strings length n (Eq. (4)), and **e** visualization of the longest atomic strings ($n = 3$ and 4) in LGPS.

Fig. 8 | Crystal structure and Li sites in LPSB.

Crystal structures of the unit cell for the cubic LPSB unit cell, illustrating the Li sites by green volumes (right). Representation of one pseudo-octahedral cage (left) in LPSB.



Even though fast-ion SSEs are generally thought to enhance ion transport through concerted diffusion mechanisms⁷, our observations indicate that, in ordered LPSB, Li atoms remain confined within the octahedral cages despite the presence of concerted jumps of up to 4 Li atoms. As a result, concerted motion does not make a significant contribution to the system's overall IC. In this case, intercage jumps must play a more decisive role in allowing fast ion transport in Li argyrodites, as recently suggested elsewhere⁵⁴.

In disordered LPSB, CrySF similarly identified 24 type 0 sites and no T2 sites were found. Key differences between ordered and disordered LPSB observed in our study include a 4% increase in average site amplitude, a 4% increase in average occupancy, and a 55% increase in the total number of jumps in the disordered system. These results align with the well-documented effects of anion disorder (S/X substitution) in halide Li-argyrodites^{52,56–58}.

In the disordered LPSB system, three Li atoms were observed to visit more than six sites, confirming the occurrence of intercage jumps (Fig. 10a). The number of reverse jumps increased by approximately 22% in the disordered system compared to the ordered one ($J^R = 264$ for ordered and $J^R = 323$ for disordered LPSB), as is also reflected by the α_k values (Fig. 10b). The collective behavior, as indicated by the number of atoms involved in simultaneous jumps and their f_n , remains similar to those in the ordered system (cf. Fig. 9c, d and Fig. 10c, d). These results are consistent with the view that anion disorder induces distortions in the lithium cages, enhancing site connectivity between adjacent cages^{52,58}. The structural distortions in the disordered system may also contribute to the increased number of reverse jumps (intracage) observed in disordered LPSB, which in turn may help the system access configurations that facilitate the creation of intercage strings with $n = 3$ and $n = 4$, connecting two cages.

By tracking the Li atoms that CrySF detected to visit more than 6 sites, we observed cages with seven Li atoms, as illustrated in Fig. 10e left (red circle). Concomitantly, cages with a deficiency of Li atoms emerge (Fig. 10e left (blue circle)). This scenario is possible thanks to the temporary allocation of Li atoms within interstitial regions of the cage with an excess of Li ions. Figure 10e (middle) shows that, to return to the equilibrium configuration (6 atoms per cage), a transition state consisting of an intercage jump within a string of length $n = 3$ occurs. This string length is explained by the sequence of movements: the first Li atom in the string moves toward an empty site in the Li-deficient cage, while the last Li atom in the string moves from an interstitial region in the Li-excess cage. In the final stage, both cages stabilize with six Li atoms each, as shown in Fig. 10e (right).

Intercage jumps in disordered LPSX have been studied by Morgan et al.⁵² through careful examination of AIMD trajectories. We have shown that this type of analysis could be simplified and further enhanced through

the tools incorporated in the CrySF code, which has revealed that intercage jumps require the formation of two types of defects: Li-deficient and Li-oversaturated cages.

Additional structures

To further validate the method, we extended the study to include five additional materials, with a primary focus on site identification and classification analysis. These structures were selected from the AIMD database <https://superionic.upc.edu/>^{11,42} and from the work of Lee et al.⁵⁹.

The results in Table 1 show strong agreement between the site analysis performed using CrySF and experimental data. CrySF successfully identified all sites and site types in the analyzed structures. Notably, in the case of β -Li₃PS₄, CrySF was able to identify even the three types of sites (8d, 4d, and 4c); however, the occupancy of the 4c sites is lower than expected. This low value is likely a consequence of the trajectory not being long enough, leading to insufficient sampling of these sites during the simulation.

Discussion

In this work, we introduced a novel unsupervised method for combined structural and diffusion analysis of atomistic trajectories, enabling the accurate and efficient characterization of mobile ion behavior in complex crystalline structures. Our method and its implementation in the CrySF code are both robust and versatile, requiring few tuning parameters, and are applicable across a wide range of materials. Through its application to well-known SSE materials such as tetragonal and cubic Li₇La₃Zr₂O₁₂ (LLZO), Li₁₀GeP₂S₁₂ (LGPS), and both ordered and disordered Li₆PS₅Br (LPSB), we demonstrated the method's effectiveness in capturing critical structural aspects and ion transport properties of these materials, as well as its potential for guiding its users towards important phenomena that may otherwise be missed in typical post-processing routines.

CrySF successfully detected all sites and site types in tetragonal LLZO. In superionic conductors, the code may merge adjacent sites with very high jumping rates, as observed for the 96h sites in cubic LLZO (see Fig. 3b), the 16h sites in LGPS (see Fig. 6), and the 48h/24g/48h (see T5-T5a-T5 in Fig. 8) sites in LPSB. We note that CrySF results for a specific material should not be generalized to other compositions in materials of the same family. For instance, aliovalent substitution in argyrodites may transform high-frequency transitions into slower, rate-limiting steps⁶⁰.

Importantly, high-frequency transitions that obfuscate site identification are not the rate-limiting steps in ionic transport. Thus, they have no significant impact on the overall IC. Taking this into consideration, CrySF was able to identify nearly all diffusionally

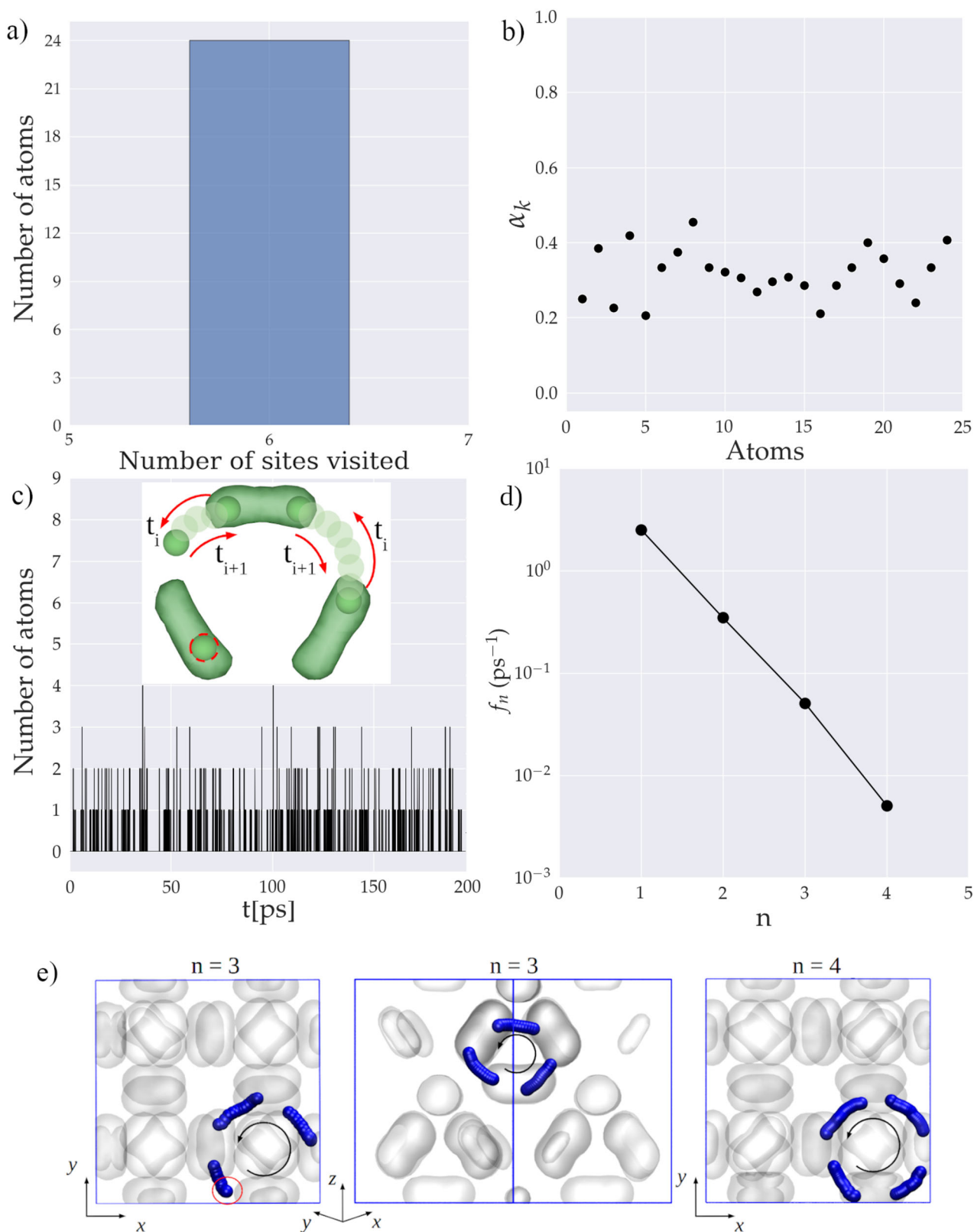


Fig. 9 | Li dynamics and diffusion strings in ordered LPSB according to CrySF. **a** Number of distinct sites visited. **b** Jump coefficient α_k (Eq. (2)). **c** Number of atoms jumping simultaneously within $\Delta t_j = 0.06$ ps. The inset illustrates simultaneous jumps of two Li atoms while a third Li atom (dashed red line) remains stationary.

Consecutive jumps are denoted by t_i and t_{i+1} in the inset. **d** String frequency, f_n , as a function of strings length n (Eq. (4)), and **e** visualization of the longest atomic strings ($n = 3$ and 4) in ordered LPSB.

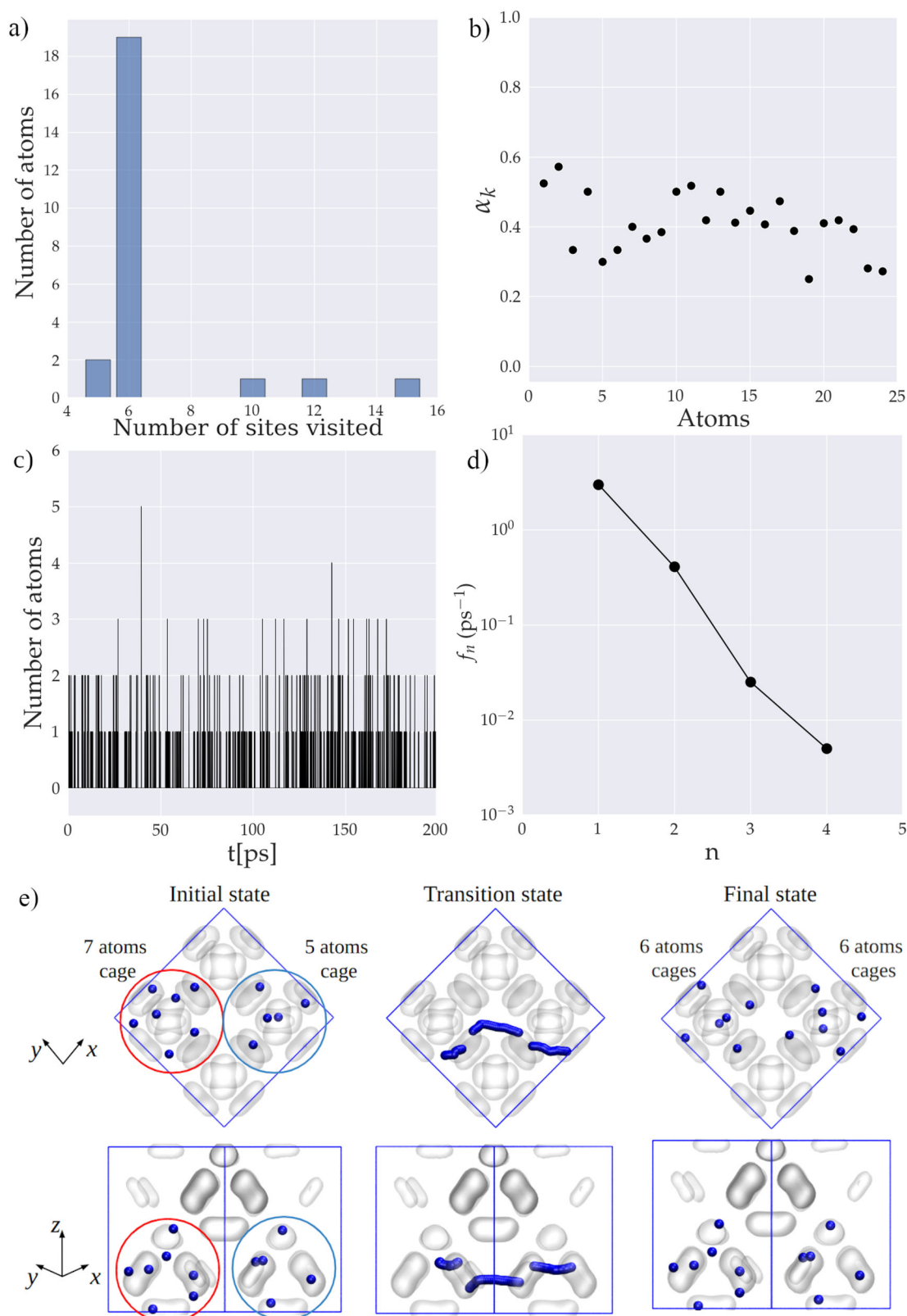


Fig. 10 | Li dynamics and diffusion strings in disordered LPSB according to CrySF. **a** Number of distinct sites visited. **b** Jump coefficient α_k (Eq. (2)). **c** Number of atoms jumping simultaneously within $\Delta t_j = 0.06$ ps. **d** String frequency,

f_n , as a function of strings length n (Eq. (4)), and **e** visualization of the longest atomic strings ($n = 3$ and 4) in disordered LPSB.

relevant sites in these superionic conductors. The only exception was the T2-type sites in LPSB, which are rarely visited and may require significantly longer trajectories. In comparison, the method developed by Kozinsky et al.¹⁹, while rather robust, can only distinguish

between some of the 96h sites in cubic LLZO at a greater computational cost.

Long strings of concerted atomic migration have traditionally been regarded as a hallmark of superionic conductors. However, our

Table 1 | Comparison between the site analysis performed by CrySF and the experimental site data from previous studies

Structure	Method	Number of sites	Site types	Occupancy	NN
Li ₃ OCl	CrySF	3	1	0.96	8
	Exp. ⁶⁵	3	3d	1.0	8
LiGaO ₂	CrySF	4	1	0.99	4
	Exp. ⁶⁶	4	4a	1.0	4
LiMn(HCO ₂) ₃	CrySF	4	1	0.99	6
	Exp. ⁶⁷	4	4a	1.0	4
Li ₂ B ₁₂ H ₁₂	CrySF	8	1	0.85	7
	Exp. ⁶⁸	8	8c	1.0	7
β -Li ₃ PS ₄	CrySF	16	3	0.85, 0.75, 0.11	4, 4, 3
	Exp. ⁶⁹	16	8d, 4d, 4c	1.0, 0.66, 0.34	4, 4, 3

analysis of Li-ion diffusion using CrySF suggests that the mere presence of these strings does not necessarily correlate with high IC. For instance, we observed strings as long as $n = 12$ in the tetragonal phase of LLZO, yet the IC remained low. Similarly, in ordered LPSB, strings of $n = 4$ were present, but no significant long-range diffusion was observed due to the confinement of Li atoms within well-defined cages. Therefore, string length is not a definitive measure of efficient ionic transport. Rather, the frequency of such occurrences, the number of distinct sites visited, and the persistence of motion in a particular direction must all be considered. This observation highlights the importance of complementary analyses, such as those integrated into the CrySF code, which provide a more comprehensive understanding of ion dynamics in crystalline materials. Such multifaceted approaches are crucial for unraveling the complex mechanisms that govern ion mobility and designing strategies to enhance it.

Methods

Crystallographic site characterization

Once the crystallographic sites are defined through the scheme presented in Fig. 1, important features of any i -th site can be readily extracted:

- Site amplitude: defined for each C_i^s as

$$A_{C_i^s} = \max_{v_v \in C_i^s} \| \mathbf{r}_{v_v} - \mathbf{r}_{C_i} \|, \quad (5)$$

where $\mathbf{r}_{v_v}$ represents the position vector of the voxels within the cluster C_i^s , and $\mathbf{r}_{C_i}$ is the position vector of the cluster centroid.

- Site volume: for each C_i^s , it is calculated as

$$V_{C_i^s} = V_{\text{voxel}} \times n_v^i, \quad (6)$$

where n_v^i is the number of voxels in cluster C_i^s and V_{voxel} is the voxel volume.

- Nearest neighbors (NN) analysis: determined by measuring the distance to the first NN (1NN) from every site of type j . The resulting values are averaged over all sites of type j and used as a cutoff distance to count the number of sites surrounding each site type j . This procedure guarantees a specific cutoff per site type.
- Site occupancy (O_i^s): It is determined by summing the density of all the voxels within a cluster C_i^s , that is:

$$O_i^s = V_{\text{voxel}} \times \sum_{l=1}^{n_v^i} \rho_l, \quad (7)$$

where ρ_l is the density in voxel l . The occupancy for site type j , O_j , is obtained by averaging O_i^s over all sites of type j .

CMD simulations

CMD simulations for the tetragonal and cubic phases of Li₇La₃Zr₂O₁₂ (LLZO) were performed using LAMMPS⁶¹. Periodic 8-unit cells ($2 \times 2 \times 2$) supercell containing 1536 atoms was used, with a time step of 2 fs. The interatomic potential was calculated as the sum of Coulombic and Buckingham interactions, as detailed in our recent work²². Both systems were equilibrated at 300 K in the isothermal-isobaric (NPT) ensemble for 1 ns, utilizing a Nose-Hoover thermostat and barostat with damping parameters of 0.2 ps for temperature and 2 ps for pressure. Since the cubic phase is not stable at 300 K, the cell was constrained to a cubic lattice shape. Production runs were then carried out in the canonical (NVT) ensemble at 300 K, lasting 15 ns for the tetragonal phase and 5 ns for the cubic phase.

AIMD simulations

AIMD simulations based on DFT were performed using the Perdew-Burke-Ernzerhof (PBE) functional⁶² as implemented in the VASP code (version 6.3)⁶³, with a cutoff energy of 520 eV. Trajectories, 200 ps long, were generated in the NVT ensemble with a Nose-Hoover thermostat at 650 K and a time step of 1.5 fs. Periodic simulation cells containing 50 atoms for Li₁₀GeP₂S₁₂ (LGPS) and 52 atoms for Li₃PS₃Br (LPSB) were employed. In the case of LPSB, two distinct S/Br site configurations were considered: an ordered scenario, with S wholly occupying the 4c sites and Br solely occupying the 4a sites, and a disordered scenario, where half of the Br atoms occupy both the 4c and 4a sites. The selected volumes were determined at zero temperature; hence, thermal expansion effects were neglected. Nevertheless, based on previously reported AIMD simulations, thermal expansion effects are not expected to significantly affect the estimation of ion-transport features at moderate temperatures⁶⁴.

To further validate our method, we applied it to the set of additional ion conducting materials, including Li₃OCl, LiGaO₂, LiMn(HCO₂)₃, Li₂B₁₂H₁₂, simulated at $T = 1250, 700, 500$, and 600 K, which are included in the AIMD database available at <https://superionic.upc.edu/>. Additional details about the simulation settings are described elsewhere^{11,42}. Finally, we used AIMD trajectories of β -Li₃PS₄ at 650 K, produced in a recent study discussing the paddlewheel effect in superionic conductors⁵⁹.

Data availability

All data supporting the findings of this study are available from the authors upon reasonable request. CMD trajectories for the tetragonal and cubic LLZO systems, as well as AIMD trajectories for LGPS and both ordered and disordered LPSB systems, are available on Zenodo (<https://doi.org/10.5281/zenodo.16910079>).

Code availability

The unsupervised method for identifying and classifying crystallographic sites in crystalline ionic conductors, as well as the tools for structural and diffusion analysis, are freely available in the open-source Python code CrySF <https://gitlab.bcamath.org/hacortes/crysf>. The code repository includes instructions and examples of use. More information about the code can be found at <https://crysf.readthedocs.io/en/latest/#installation>.

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Author contributions

H.A.C. conceptualized the study, developed the methodology, conducted the investigation, and contributed to the writing of the original draft as well as the review and editing of the manuscript. M.R.B. and E.A. contributed to conceptualization, investigation, supervision, funding acquisition, and the review and editing of the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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